

UNIVERSITY OF SOUTHERN CALIFORNIA

EE 483: Digital Signal Processing

Lecture 10: DFT Matrix, DCT, and Transforms as Representations

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1 DFT in Matrix Form

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn}, \quad 0 \leq k \leq N-1 \quad \Leftrightarrow \quad \vec{X} = \mathbf{D}_N \vec{x},$$

$$\begin{bmatrix} X[0] \\ X[1] \\ X[2] \\ \vdots \\ X[N-1] \end{bmatrix} = \underbrace{\begin{bmatrix} W_N^0 & W_N^0 & W_N^0 & \cdots & W_N^0 \\ W_N^0 & W_N^1 & W_N^2 & \cdots & W_N^{N-1} \\ W_N^0 & W_N^2 & W_N^4 & \cdots & W_N^{2(N-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ W_N^0 & W_N^{N-1} & W_N^{2(N-1)} & \cdots & W_N^{(N-1)^2} \end{bmatrix}}_{\mathbf{D}_N, [\mathbf{D}_N]_{kn} = W_N^{kn}} \begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ \vdots \\ x[N-1] \end{bmatrix}.$$

Claim. $\mathbf{D}_N$ and $\mathbf{D}_N^{-1}$ are related:

$$\mathbf{D}_N^{-1} = \frac{1}{N} \mathbf{D}_N^H \quad \Rightarrow \quad \frac{1}{\sqrt{N}} \mathbf{D}_N \text{ is a **unitary** matrix (after scaling).}$$

A unitary matrix $\mathbf{U}$ satisfies $\mathbf{U}^{-1} = \mathbf{U}^H$.

Remark

(addendum to orig lecture note).

The claim is the inverse DFT of Lecture 8 in matrix language: $[\mathbf{D}_N^H]_{nk} = W_N^{-kn}$, and

$$[\mathbf{D}_N^H \mathbf{D}_N]_{\ell k} = \sum_{n=0}^{N-1} W_N^{-\ell n} W_N^{kn} = N \delta[\langle k - \ell \rangle_N] \quad \Rightarrow \quad \mathbf{D}_N^H \mathbf{D}_N = N \mathbf{I}.$$

Property (energy preservation).

For unitary $\mathbf{U}$ and $\vec{X} \triangleq \mathbf{U}\vec{x}$,

$$\begin{aligned} \sum_{n=0}^{N-1} |x[n]|^2 &= \vec{x}^H \vec{x} = \vec{x}^H \mathbf{U}^{-1} \mathbf{U} \vec{x} = \vec{x}^H \mathbf{U}^H \mathbf{U} \vec{x} = (\mathbf{U}\vec{x})^H \mathbf{U}\vec{x} \\ &= \vec{X}^H \vec{X} = \sum_{k=0}^{N-1} |X[k]|^2. \end{aligned}$$

With $\mathbf{U} = \frac{1}{\sqrt{N}} \mathbf{D}_N$ this is exactly Parseval's theorem for the DFT, $\sum |x[n]|^2 = \frac{1}{N} \sum |X[k]|^2$ (Lecture 8).

2 1.1 Unitary Transforms

- Unitary matrices/transforms **preserve signal energy** (Parseval's theorem)
 $\Rightarrow$ good for approximation.

Inverse of a unitary transform.

If

$$\mathcal{T}\{x[n]\}[k] = \sum_{n=0}^{N-1} c[n, k] x[n]$$

and $\mathcal{T}\{\cdot\}$ is unitary, then

$$\mathcal{T}^{-1}\{X[k]\}[n] = \sum_{k=0}^{N-1} c^*[n, k] X[k].$$

For the DFT, $c[n, k] = W_N^{nk}$ and $c^*[n, k] = W_N^{-nk}$ (we still need the factor $\frac{1}{N}$, since

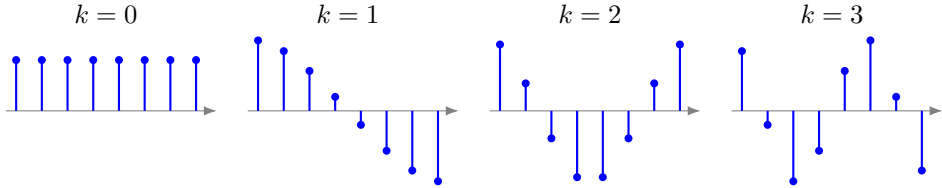
$\mathbf{D}_N$ itself is unitary only after scaling by $\frac{1}{\sqrt{N}}$).

2 Discrete Cosine Transform (DCT)

DCT (one version)¹.

$$X[k] = w[k] \sum_{n=0}^{N-1} x[n] \cos\left(\frac{\pi(2n+1)k}{2N}\right), \quad w[k] = \begin{cases} \frac{1}{\sqrt{N}}, & k = 0, \\ \sqrt{\frac{2}{N}}, & 1 \leq k \leq N-1. \end{cases}$$

- Like a DFT of a **symmetric length- $2N$ sequence** (shown below).
- Real sequences are mapped to real sequences.
- Used in compression (JPEG, MPEG, MP3, ...).
- Can also be used for denoising.



First four DCT-II basis vectors $\cos(\frac{\pi(2n+1)k}{2N})$ for $N = 8$ (scaled by $w[k]\sqrt{N/2}$), $n = 0, \dots, 7$.

3 2.1 DCT, KLT, and the Inverse DCT

- The DCT approximates the **Karhunen–Loève Transform (KLT)** for a 1st-order Markov chain (EE 562).
- The KLT is a **decorrelating** transform.

¹This is the orthonormal **DCT-II**, the version computed by MATLAB's `dct`; other DCT types (I, III, IV) differ in how the length- N block is symmetrically extended.

Inverse DCT (unitary).

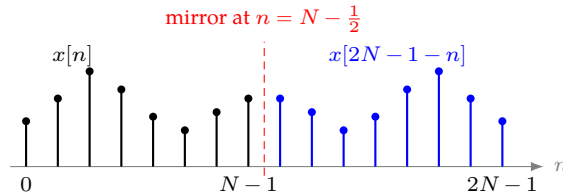
$$x[n] = \sum_{k=0}^{N-1} w[k] \cos\left(\frac{\pi(2n+1)k}{2N}\right) X[k].$$

This is the unitary-inverse rule above with $c[n, k] = w[k] \cos\left(\frac{\pi(2n+1)k}{2N}\right)$ real, so $c^* = c$.

2.2 Why “a DFT of a Symmetric Length- $2N$ Sequence”

Consider the symmetric extension

$$\tilde{x}[n] = \begin{cases} x[n], & 0 \leq n \leq N-1, \\ x[2N-1-n], & N \leq n \leq 2N-1. \end{cases}$$



Its $2N$ -point DFT is

$$\begin{aligned} \tilde{X}[k] &= \sum_{n=0}^{N-1} x[n] W_{2N}^{nk} + \sum_{n=N}^{2N-1} x[2N-1-n] W_{2N}^{nk} \\ &= \sum_{n=0}^{N-1} x[n] \underbrace{\left[W_{2N}^{nk} + W_{2N}^{-(n+1)k} \right]}_{\text{almost a cosine}} \end{aligned}$$

(in the second sum substitute $m = 2N - 1 - n$ and use $W_{2N}^{2Nk} = 1$). Multiplying by the half-sample phase $W_{2N}^{k/2}$ centers the pair:

$$\tilde{X}[k] W_{2N}^{k/2} = \sum_{n=0}^{N-1} x[n] \underbrace{\left[W_{2N}^{(n+\frac{1}{2})k} + W_{2N}^{-(n+\frac{1}{2})k} \right]}_{2 \cos\left(\frac{2\pi(n+\frac{1}{2})k}{2N}\right)} = 2 \sum_{n=0}^{N-1} x[n] \cos\left(\frac{\pi(2n+1)k}{2N}\right).$$

The scaling term $w[k]$ ensures that the transform is **unitary**.

Remark

(addendum to orig lecture note).

So, for $0 \leq k \leq N - 1$, $X_{\text{DCT}}[k] = \frac{w[k]}{2} W_{2N}^{k/2} \tilde{X}[k]$: the DCT can be computed with one $2N$ -point FFT. Because the extension $\tilde{x}$ has no jump at the block boundary (unlike the implicit periodic extension of the DFT), its spectrum decays faster — one reason the DCT compacts energy into few coefficients and suits compression.

4 3 2D Transforms (Separable)

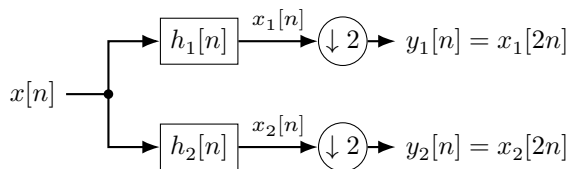
Input $x[m, n]$, $0 \leq m, n \leq N - 1$. The **2D DFT** is

$$X[k, q] = \sum_{m=0}^{N-1} \underbrace{\left[\sum_{n=0}^{N-1} x[m, n] W_N^{nq} \right]}_{\text{1D DFT of each row}} W_N^{mk}.$$

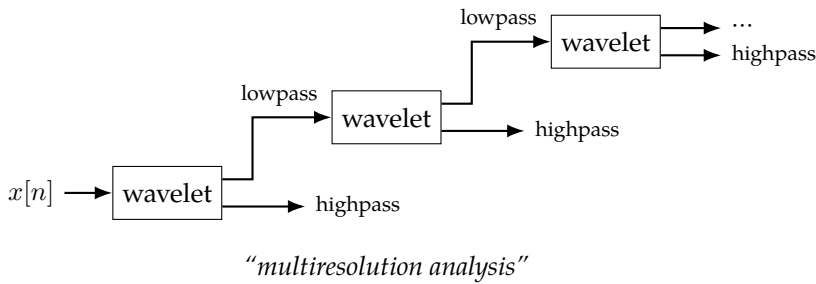
The outer sum is then a **1D DFT of each column** of the row-transformed array. (Separable: N row DFTs followed by N column DFTs.)

4 Wavelet Transforms (EE 596)

Discrete, unitary. One level splits the signal into a lowpass and a highpass branch, each downsampled by 2:



with $h_1[n]$ = **low-pass** filter and $h_2[n]$ = **high-pass** filter. Repeating on the low-pass branch gives a **multiresolution analysis**:



5 Transforms as Representations

We may be used to thinking of transforms as **operations** that we apply to signals to gain insight. But it can also be useful to think of them as **representations** of the signal in a **new basis**, e.g.,

$$x[n] = \sum_k X[k] \underbrace{\varphi_k[n]}_{\text{basis function}}.$$

The **inversion formula** tells you what the φ_k are:

Examples.

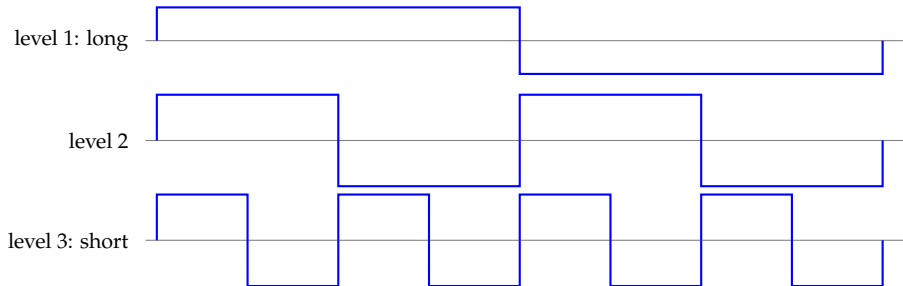
- **DFT:** the basis functions are complex exponentials, $\varphi_k[n] = \frac{1}{N} e^{+j\frac{2\pi kn}{N}}$.
- **DCT:** the basis functions are cosines, $\varphi_k[n] = w[k] \cos\left(\frac{\pi(2n+1)k}{2N}\right)$.
- **“Identity transform”:** the basis functions are shifted impulses, $\varphi_k[n] = \delta[n - k]$.

5.1 Wavelet Bases

Wavelet basis functions have **different extents at different decomposition levels**:

- shifting & stretching some original function, $\varphi_k[n] \sim f\left(\frac{n - m_k}{\alpha}\right)$;
- fractal “**self-similarity**.”
- **“Haar”** wavelets are good for approximating **piecewise constant** signals.

- Other families (e.g., “**Daubechies**” wavelets) are good at approximating more complicated signals (like **piecewise polynomials**).



Haar basis functions (illustration, not in orig lecture note): each level shifts and compresses the same “up-down” shape.

Readings

- Sanjit K. Mitra, *Digital Signal Processing: A Computer-Based Approach*, 4th ed., Ch. 5, 3.8
- Monson H. Hayes, *Schaum's Outline of Digital Signal Processing*, 2nd ed., Ch. 3